The Rendering Universe: Causal Refinement as a Unified Framework for Cosmology

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Abstract

We propose that cosmic expansion is not spatial stretching but **causal refinement**: the dynamical increase of resolution $\mathcal{R}(t)$, measured in Planck pixels per side of a fixed Minkowski canvas of size 1. We present this as a **phenomenological effective framework** whose macro-dynamics are consistent with several independent datasets. We show that (1) $\rho_{\text{vac}} = M_{\text{Pl}}^4/\mathcal{R}^4$ resolves the 120-order vacuum catastrophe up to a factor ~ 40 (as an IR/holographic regularization of gravitating vacuum energy), (2) a direct $\mathcal{R} \propto t^{\beta}$ likelihood against the official DESI DR2 BAO data (13 points, full covariance) gives $\beta_0 = 0.055^{+0.045}_{-0.050}$, resolving an earlier estimator tension, though BAO alone prefer this over ΛCDM by only $\Delta\chi^2 \approx 1$, (3) a pilot SPARC comparison shows a two-parameter cored profile fits better than NFW in 7/8 galaxies (though cored profiles outperform NFW independently of theory and the profile is not yet derived from the link-counting Postulate 6), (4) variable local $\mathcal{R}$ can account for the large stacked ISW anomaly while consistent with cross-correlations, and (5) JWST high-z stellar mass densities are compatible with resolution sub-sampling. The framework makes a sharp prediction: $\Delta T \approx 14$ μK for DESI×Planck void stacking at $z \sim 0.3$ (vs ΛCDM ~ 2 μK). A first-principles derivation of $\mathcal{R}(t, \mathbf{x})$ from causal-set dynamics and full statistical analyses are required next steps.

1 Introduction

The standard ΛCDM model successfully describes a wide range of cosmological observations but relies on several unexplained components: a cosmological constant fine-tuned to 10^{-120} in Planck units (Weinberg, 1989), cold dark matter particles undetected after decades of searches (Bertone and Hooper, 2018), and an inflationary epoch driven by an unknown field (Guth, 1981). The recent DESI DR2 results (DESI Collaboration, 2025) showing 3.1–4.2σ preference for evolving dark energy ($w_0 > -1$, $w_a < 0$) add further tension.

We propose a simpler ontological shift: **the universe is not expanding – it is being rendered at increasing resolution.** This single hypothesis, inspired by the causal set approach to quantum gravity (Bombelli et al., 1987; Sorkin, 2003), replaces the three ad-hoc components with one mechanism.

2 Theoretical Framework

2.1 The Central Postulate

Postulate 1 (Fixed Canvas). Spacetime, at the fundamental level, is Minkowski flat:

$$ds^2_{\text{fundamental}} = -c^2 dt^2 + dr^2 + r^2 d\Omega^2 \tag{1}$$

The universe is a manifold of total "size" 1 in suitable conformal units. It does not expand or contract.

Postulate 2 (Finite Resolution). Observations are made with finite causal resolution $\mathcal{R}(t)$, the number of Planck pixels per side:

$$\mathcal{R}(t) \in [1, \infty) \tag{2}$$

The resolution increases monotonically from $\mathcal{R} = 1$ at the Big Bang to $\mathcal{R}_0 \sim 10^{30}$ today, corresponding to $N_0 \sim \mathcal{R}_0^4 \sim 10^{120}$ causal elements (matching the holographic bound; Bousso, 2002).

Postulate 3 (Resolution Redshift). The cosmological redshift is not a stretching of space but a change in resolution between emission and reception:

$$1 + z = \frac{\mathcal{R}(t_0)}{\mathcal{R}(t_e)} \tag{3}$$

This follows directly from comparing the same physical signal sampled at two different resolutions.

2.2 Derivation from Causal Set Dynamics

The framework is grounded in causal set theory (CST; Bombelli et al., 1987). In CST, spacetime is a partially ordered set of N elements. The number of elements N scales as the spacetime volume in Planck units:

$$N \sim \frac{V}{\ell_P^4} \sim \mathcal{R}^4 \tag{4}$$

The causal set grows via **Classical Sequential Growth (CSG)** (Rideout and Sorkin, 2000). In CSG, new elements are added one at a time with transition probabilities that respect causality and discrete general covariance.

Derivation of $\mathcal{R}(t)$ dynamics. The growth rate of causal set elements in Minkowski space ($V \propto t^4$) is:

$$\frac{dN}{dt} = \frac{d}{dt}\left(\frac{V(t)}{\ell_P^4}\right) = \frac{4V(t)}{\ell_P^4\, t} \tag{5}$$

Using $N \sim \mathcal{R}^4$:

$$4\mathcal{R}^3 \frac{d\mathcal{R}}{dt} = \frac{4\mathcal{R}^4}{t} \quad \Rightarrow \quad \frac{d\mathcal{R}}{dt} = \frac{\mathcal{R}}{t} \tag{6}$$

In a universe with matter, growth is modulated by the rate of causal connections per element:

$$\frac{d\mathcal{R}}{dt} = \frac{c}{\ell_P} \left(\frac{\rho_m(t)}{\rho_c(t)} \right)^{1/2} \tag{7}$$

During matter and radiation domination, $\rho_m/\rho_c \approx 1$, recovering $\mathcal{R} \propto t$. During dark energy domination, $\rho_m/\rho_c \to 0$, slowing the refinement. The growth exponent $\beta \equiv d\ln\mathcal{R}/d\ln t$ evolves from $\beta \approx 1$ (early) to $\beta_0 \approx 0.05$–0.15 today; the current spread in β_0 reflects a real tension between two estimators, discussed openly in Section 3.

2.3 The Effective Metric and Emergent Gravity

If the fundamental metric is Minkowski but observations are made at resolution $\mathcal{R}(x)$, the effective metric emerges as a conformal rescaling:

$$g_{\mu\nu}^{(\mathrm{eff})}(x) = \eta_{\mu\nu} \frac{\ell_P^2}{\mathcal{R}(x)^2} \tag{8}$$

The Ricci curvature tensor for this conformally flat metric is:

$$\begin{aligned} R_{\mu\nu} = {} & \frac{2}{\mathcal{R}}\partial_\mu\partial_\nu\mathcal{R} - \frac{1}{\mathcal{R}}\Box\mathcal{R}\,\eta_{\mu\nu} \\ & - \frac{2}{\mathcal{R}^2}\partial_\mu\mathcal{R}\partial_\nu\mathcal{R} + \frac{1}{\mathcal{R}^2}(\partial\mathcal{R})^2\eta_{\mu\nu} \end{aligned} \tag{9}$$

The Ricci scalar is:

$$R = -\frac{6}{\mathcal{R}}\Box\mathcal{R} + \frac{6}{\mathcal{R}^2}(\partial\mathcal{R})^2 \tag{10}$$

Limit of constant $\mathcal{R}$. When $\partial_\mu\mathcal{R} = 0$, all curvature vanishes. The metric reduces to Minkowski locally.

Limit of slowly varying $\mathcal{R}$. The Einstein tensor becomes:

$$G_{\mu\nu} \approx \frac{2}{\mathcal{R}}\big(\partial_\mu\partial_\nu\mathcal{R} - \eta_{\mu\nu}\Box\mathcal{R}\big) \tag{11}$$

Identifying the effective stress-energy tensor of the resolution field:

$$T_{\mu\nu}^{(\mathcal{R})} \equiv \frac{1}{4\pi G\mathcal{R}}\big(\partial_\mu\partial_\nu\mathcal{R} - \eta_{\mu\nu}\Box\mathcal{R}\big) \tag{12}$$

The cosmological constant emerges as $\Lambda = 3(\dot{\mathcal{R}}/\mathcal{R})^2_{\mathrm{bg}}$, matching the observed value.

2.4 The Vacuum Catastrophe

The quantum vacuum energy density, integrated to infinite momentum, diverges. We postulate that the vacuum energy *that gravitates* is bounded by the coherence scale of the causal render:

$$\rho_{\mathrm{vac}} = \frac{1}{2\pi^2}\int_0^{2\pi/(\ell_P\mathcal{R})} k^2\sqrt{k^2+m^2}\,dk \approx \frac{2\pi^2 M_{\mathrm{Pl}}^4}{\mathcal{R}^4} \tag{13}$$

with effective momentum bound $k_{\max} = 2\pi M_{\text{Pl}}/\mathcal{R}$.[1] Today, $\mathcal{R}_0 \sim 10^{30}$ yields $\rho_{\text{vac}}^{\text{pred}} \approx 10^{-120}\, M_{\text{Pl}}^4$ (observed: $2.3 \times 10^{-122}\, M_{\text{Pl}}^4$), removing the 120-order discrepancy up to a factor of ~ 40.

What this ansatz is and is not. The bound $k_{\max} \sim M_{\text{Pl}}/\mathcal{R}_0 \sim 10^{-11}$ eV must *not* be read as a cutoff on laboratory quantum field theory, which is verified to TeV scales; modes above $k_{\max}$ exist and interact, but their zero-point energy is postulated not to source the effective $T_{\mu\nu}^{(\mathcal{R})}$. This is an infrared/holographic regularization of the *gravitating* vacuum energy, in the spirit of the Cohen–Kaplan–Nelson bound $\rho_{\text{vac}} \lesssim M_{\text{Pl}}^2 L^{-2}$ for a region of size L (Cohen et al., 1999) and of holographic entropy bounds (Bousso, 2002): taking $L \sim R_H$ gives $\rho \sim M_{\text{Pl}}^2 H_0^2 \sim 10^{-122}\, M_{\text{Pl}}^4$, numerically consistent with our $\mathcal{R}^{-4}$ scaling. A first-principles derivation of which modes gravitate – presumably from the causal-set structure, cf. the everpresent-Λ program (Sorkin, 2005; Ahmed et al., 2004) – is an open problem of the framework; Eq. (13) is an ansatz whose independent support is the DESI-facing dynamics of Section 4, not a derivation from quantum field theory.

2.5 Temperature as Sampling Noise

The CMB temperature of $T_{\text{CMB}} = 2.725$ K emerges naturally from the sampling noise of a discrete causal set. A first-principles derivation would require the full causal set dynamics; here we present only an empirical scaling ansatz. The observed temperature is reproduced by

$$T_{\text{CMB}} \sim \frac{T_{\text{Planck}}}{\mathcal{R}_0} \cdot \frac{1}{50} \tag{14}$$

with $T_{\text{Planck}} = \sqrt{\hbar c^5/(G k_B^2)} \approx 1.4 \times 10^{32}$ K and $\mathcal{R}_0 \sim 10^{30}$, giving $T_{\text{Planck}}/\mathcal{R}_0 \approx 140$ K, reduced by an empirical factor $\sim 1/50$ to match 2.725 K.[2] Deriving the exact prefactor from first principles is deferred to a dedicated treatment.

3 Dark Energy as Quantization Noise

3.1 Equation of State

Since $\rho_{\text{DE}} \propto \mathcal{R}^{-4}$, the equation of state is:

$$w(a) = -1 + \frac{4}{3}\frac{d\ln\mathcal{R}}{d\ln a} \tag{15}$$

This follows from $w = -1 + (1/3)\,d\ln\rho/d\ln a$ and $\rho_{\text{DE}} \propto \mathcal{R}^{-4}$. Converting the time-derivative of $\mathcal{R}$ to a scale-factor derivative requires an explicit $a(t)$ relation. In the matter-dominated era $a \propto t^{2/3}$, giving $d\ln\mathcal{R}/d\ln a = (3/2)\,d\ln\mathcal{R}/d\ln t \equiv (3/2)\,\beta$. With $\mathcal{R} \propto t^{\beta}$:

$$w_0 = -1 + 2\beta_0 \tag{16}$$

[1] The factor 2π is absorbed into the definition of $\mathcal{R}$ in subsequent sections to keep notation simple.

[2] This is a purely empirical calibration; the factor 1/50 is not derived from first principles. A complete treatment would compute the mode-counting on the 3-dimensional CMB surface, for which the 4-dimensional counting $T \sim T_{\text{Planck}}/\mathcal{R}_0^2 \sim 10^{-28}$ K is far too small, suggesting that the relevant degrees of freedom are those of the surface rather than the bulk. This remains open for future work.

An earlier version of this section reported a factor ∼2–3 tension between two estimates of β_0: per-bin reconstruction of $\mathcal{R}(z)$ from the BAO distances gave $\beta_{\rm eff} \approx 0.02$–$0.07$, while mapping the published DESI w_0w_aCDM fit ($w_0 = -0.727 \pm 0.067$; DESI Collaboration, 2025) through $w_0 = -1 + 2\beta_0$ suggested $\beta_0 \approx 0.14$. We have now performed the direct likelihood fit that we had identified as the arbiter (repository experiment 15): the self-consistent model $\mathcal{R} \propto t^\beta$, $\rho_{\rm DE} \propto \mathcal{R}^{-4}$, fit to the official DESI DR2 release of 13 BAO measurements (D_V, D_M, D_H over r_d) with the full published covariance, profiling over Ω_m and H_0r_d. The result is $\beta_0 = 0.055^{+0.045}_{-0.050}$ (68%, BAO alone; $0.040^{+0.040}_{-0.040}$ with a Planck-like Ω_m prior), with excellent goodness of fit ($\chi^2/\text{dof} = 0.92$). The reconstruction band lies inside the 68% interval, while $\beta_0 = 0.14$ is excluded at 95% once the Ω_m prior is applied: the apparent tension was an artifact of translating a different model (w_0w_aCDM) fit to a different data combination (BAO+CMB+SNe) through the matter-era conversion. Two honest caveats follow. First, BAO alone prefer $\beta > 0$ over ΛCDM ($\beta = 0$, nested) by only $\Delta\chi^2 \approx 1$: the framework is *consistent with*, but not yet *preferred by*, these data. Second, the exact present-day conversion is $w_0 = -1 + (4/3)\,\beta_0/(H_0t_0)$, giving $w_0^{\rm eff} \approx -0.92$ at the best fit; the combined BAO+SNe likelihood required by falsifier 4 (Section 8.3) remains the outstanding test. **Caveat:** The conversion $a \propto t^{2/3}$ is standard FRW. In the strict Render framework, $1+z = \mathcal{R}_0/\mathcal{R}(t)$ replaces a as the expansion parameter, and the relation $\mathcal{R} \propto t^\beta$ is the fundamental one. The w parameterization is used here only to connect to DESI observations; a more fundamental prediction is $\mathcal{R}(t)$ directly.

3.2 DESI DR2 Comparison

DESI DR2 (DESI Collaboration, 2025) measured BAO from 14 million galaxies ($0.1 < z < 4.2$). The combination DESI+CMB+SNe prefers evolving dark energy at 2.8–4.2σ, with $w_0 > -1$ and $w_a < 0$. In the present framework this is interpreted phenomenologically: the strict relation $1+z = \mathcal{R}_0/\mathcal{R}$ gives the leading sampling-redshift map, while the effective DESI equation of state constrains the time-dependence of the refinement rate $\beta = d\ln\mathcal{R}/d\ln t$. A full likelihood analysis should fit $\mathcal{R}(t)$ directly rather than infer it through a standard w_0–w_a parameterization.

4 Dark Matter as Causal Aliasing

Postulate 5 (Causal Aliasing). What is called "dark matter" is a causal aliasing effect: the residual correlation of the primordial causal pixel that seeded a galaxy.

4.1 Velocity Profile

The causal halo's velocity profile:

$$V^2_{\rm halo}(R) = V^2_{\rm flat}\frac{(R/R_c)^2}{1 + (R/R_c)^2} \tag{17}$$

For $R \ll R_c$: $V_{\rm halo} \propto R$; for $R \gg R_c$: $V_{\rm halo} \to V_{\rm flat}$.

4.2 SPARC Test

As a pilot test against the SPARC database (Lelli et al., 2016), we fit eight representative galaxies with published rotation-curve decompositions:

Table 1: Causal halo vs NFW fit for SPARC galaxies. Full 8-galaxy test sample.

Galaxy	N_{pts}	V_{flat}	R_c	χ^2_{causal}	χ^2_{NFW}	Winner
DDO154	12	50	3.2	1.10	10.10	Causal
NGC2403	45	150	12.6	3.34	16.88	Causal
NGC2841	49	235	9.9	0.67	2.15	Causal
NGC3198	33	140	40.0	25.36	41.77	Causal
NGC7331	14	125	40.0	19.46	23.78	Causal
UGC09133	35	205	40.0	4.77	8.59	Causal
NGC5907	19	175	40.0	12.80	19.78	Causal
UGC02885	16	250	40.0	5.52	3.92	NFW

The causal profile wins in 7 of 8 test galaxies. The single NFW winner (UGC02885) has a large R_c indicating the causal halo has not yet transitioned to flat within the data range. This is a small test sample – a full SPARC analysis with profile decomposition and Bayesian model comparison is deferred to future work.

5 Cosmic Structure from Resolution

Postulate 6 (Inhomogeneous Resolution via Link Counting). The spatial variation of the resolution field is sourced by matter through the causal *link* structure of the underlying causal set. Writing $\psi(x) \equiv \ln[\mathcal{R}(x)/\bar{\mathcal{R}}(t)]$, we define

$$\psi(x) = \mu\, K \sum_{s\,\in\,\text{matter}} L(s,x), \qquad K = \frac{1}{2\pi}\sqrt{\frac{\rho_c}{6}}, \tag{18}$$

where $L(s,x) = 1$ if (s,x) is a link (a causal relation with no intermediate element), K is the normalization of the discrete retarded propagator of Johnston (Johnston, 2008), ρ_c is the sprinkling density, and μ a coupling constant. This definition is intrinsically non-local – links concentrate on the past light cone (Myrheim, 1978; Bombelli et al., 1987) – and inherits the program of recovering geometry from causal order and counting (Benincasa and Dowker, 2010).

Two properties make this the correct formulation. First, numerical sprinkling experiments (repository experiments 12–13) show that the link-counted ψ reproduces the Newtonian kernel quantitatively: potential $\propto 1/r$ (measured exponent -0.984 ± 0.031), force $\propto 1/r^2$ (-2.03 ± 0.23), static limit, and Johnston normalization verified to 1–3% with no fitted parameters. Second, the same experiments show that the naive alternative – counting all elements of the causal past interior – fails: it has no static limit and yields a distance-independent force ($1/r^2$ excluded at $> 500\sigma$), and is therefore *discarded* as a definition of $\mathcal{R}$. Since $\psi > 0$ near mass, $\mathcal{R}$ is locally enhanced and proper time $d\tau = (\ell_P/\mathcal{R})\, dt$ runs slow near mass, with the correct sign for gravitational redshift. An earlier local form of this postulate, $\partial\mathcal{R}/\partial t \propto \rho(\mathbf{x},t)/\rho_c(t)$, is retained only as the homogeneous background growth law; as a local sourcing rule it cannot reproduce potential-dependent effects (e.g. tower gravitational redshift experiments, where local density is nearly equal at both heights) and is superseded by Eq. (18).

5.1 JWST: Massive Galaxies at High Redshift

The stellar mass density (Labbé et al., 2023; Harvey et al., 2024; Boylan-Kolchin, 2023) follows from the resolution scaling. From Postulate 3 ($1+z=\mathcal{R}_0/\mathcal{R}$):

$$\mathrm{SMD}(z) \approx \mathrm{SMD}(0)\left(\frac{\mathcal{R}(z)}{\mathcal{R}_0}\right)^3 = \mathrm{SMD}(0)\,(1+z)^{-3} \tag{19}$$

Table 2: JWST stellar mass density vs redshift.

z	JWST (log)	ΛCDM	This work
7.0	$10^{7.36}$	$10^{7.15}$	$10^{6.27}$
10.5	$10^{6.20}$	$10^{5.50}$	$10^{5.91}$
12.5	$10^{5.68}$	$10^{4.80}$	$10^{5.76}$

We note the crossover openly: the $(1+z)^{-3}$ sub-sampling scaling outperforms the ΛCDM expectation only at $z \gtrsim 10$ (0.3 dex vs 0.9 dex deviation at $z = 12.5$), while at $z \approx 7$ it *under*predicts the observed SMD by an order of magnitude, worse than ΛCDM. The natural reading is that Eq. (19) is at best the high-redshift limiting behavior, with standard hierarchical growth dominating by $z \sim 7$; a model combining both regimes is required before this can be claimed as evidence, and we list it as such rather than as a confirmed success.

5.2 The Integrated Sachs-Wolfe Puzzle

The ISW arises from variable $\mathcal{R}$ along the photon path (Sachs and Wolfe, 1967). The cross-correlation ISW signal is also consistent with this picture (Krolewski and Ferraro, 2022):

$$\frac{\Delta T}{T_{\mathrm{CMB}}} = \int_{\mathrm{LOS}} \frac{\dot{\mathcal{R}}}{\mathcal{R}}\frac{\delta\mathcal{R}(l)}{\bar{\mathcal{R}}}\frac{dl}{c} \tag{20}$$

Empirical calibration. Using the Hansen+ Hansen et al. (2025) measurement at $z < 0.03$:

$$\frac{\Delta T}{T_{\mathrm{CMB}}} = A\,\frac{L}{R_H}\,e^{-z/z_0}, \quad A = 0.0024,\ z_0 = 0.30 \tag{21}$$

Table 3: ISW stacking measurements vs prediction.

Experiment	z	Observed (μK)	Predicted (μK)	ΛCDM
Granett+ 2008 (Granett et al., 2008)	0.50	9.6 ± 2.2	7.0	~ 2
Kovacs+ 2021 (Kovacs et al., 2021)	0.55	8.0 ± 3.0	5.9	~ 2
Hansen+ 2025 (Hansen et al., 2025)	0.01	30 ± 9	28.7	~ 2

6 Black Holes as Local $\mathcal{R} = 1$

Postulate 7 (Black Hole Pixel). Inside a black hole horizon, the local resolution collapses to $\mathcal{R} = 1$ – a single causal pixel. The entropy $S = A/(4\ell_P^2)$ counts the elements on the horizon. Information is not lost but maximally compressed; Hawking radiation decompresses it over the evaporation time.

7 Prediction: DESI $\times$ Planck Void Stacking

DESI DR2 contains ~ 14 million galaxies at $0.1 < z < 2.1$. Stacking voids at $z \sim 0.3$:

Table 4: ISW prediction for DESI voids.

z	This work (μK)	ΛCDM (μK)
0.1	27	~ 2
0.3	14	~ 2
0.5	7	~ 2

At $z \sim 0.3$, the prediction is $\sim 7\times$ larger than ΛCDM.

8 Discussion

8.1 Relation to Existing Alternatives

Causal set theory provides the closest technical precedent for this proposal: spacetime discreteness is encoded by a partial order, the number of elements estimates spacetime volume, and classical sequential growth gives a natural language for causal refinement (Bombelli et al., 1987; Rideout and Sorkin, 2000; Dowker, 2013). The "everpresent Λ" program similarly argues that discreteness-induced fluctuations of causal-set volume can keep an effective cosmological term near the Hubble scale without a conventional 10^{-120} fine-tuning (Sorkin, 2005; Ahmed et al., 2004). The Render framework should be read as a coarse-grained phenomenological extension of that idea: $\mathcal{R}(t, \mathbf{x})$ is an effective macroscopic variable whose dynamics remain to be derived from an action or stochastic growth law.

The initial condition $\mathcal{R} = 1$ has a precise counterpart in cyclic causal-set cosmology: a *post* – a single element through which every causal path passes – separates successive cosmic cycles in CSG models, and Sorkin's "cosmic renormalization" shows that the effective dynamical parameters are renormalized at each passage through a post, providing a mechanism for parameter naturalness without fine-tuning (Sorkin, 2000; Dowker and Zalel, 2017); observables for such cyclic models have recently been constructed (Dowker and Zalel, 2023). A speculative extension of the present framework would identify the $\mathcal{R} \to \infty$ asymptotics of one cycle with the $\mathcal{R} = 1$ post of the next, in structural analogy with the conformal crossover of Penrose's Conformal Cyclic Cosmology (Gurzadyan and Penrose, 2010). We note, however, that CCC's proposed CMB signatures (low-variance concentric circles) were not confirmed by independent reanalyses (Moss et al., 2011), and we do *not* adopt the cyclic identification here: the framework as formulated requires only a single monotonic epoch, and none of its falsifiers (Section 8.3) depends on it.

Table 5: Comparison with existing theories.

Theory	Relationship	Key difference
Causal sets (Bombelli et al., 1987; Dowker, 2013)	Foundation	$\mathcal{R}(t)$ as coarse-grained variable
Everpresent Λ (Sorkin, 2005; Ahmed et al., 2004)	$\Lambda \propto 1/\sqrt{V}$	$\mathcal{R}^4$ scaling mechanism
Emergent gravity (Verlinde, 2017)	$a_0 = cH_0$	Sampling, not entropy
MOND (Milgrom, 1983)	$a_0 = cH_0/2\pi$	a_0 as render refresh rate

8.2 Limitations

1. The dynamics of $\mathcal{R}(t)$ require a first-principles derivation from the causal set action or a stochastic growth law.

2. The transition from $\mathcal{R}$ inhomogeneity to Einstein's equations requires a full action principle.

3. **The ISW coupling constant $A = 0.0024$ is empirically calibrated.** A first-principles calculation from the causal set structure is needed.

4. The SPARC comparison is currently an eight-galaxy pilot test; a publishable claim requires the full sample, measurement errors, nuisance parameters, and Bayesian or information-criterion model comparison. Moreover, the causal-halo profile of Eq. (17) is a *cored* profile, and cored profiles are known to outperform NFW on dwarf and low-surface-brightness rotation curves independently of any specific theory (the core–cusp problem; de Blok, 2010); the 7/8 result therefore does not by itself discriminate this framework from other cored alternatives.

5. **The causal-halo profile is not derived from Postulate 6.** The link-counting definition of $\mathcal{R}$ sourced by baryons reproduces the Newtonian kernel (experiments 12–13), and Newtonian gravity from baryons alone does not produce flat rotation curves. Flat curves therefore require additional physics not yet present in the framework (e.g. inhomogeneous-sprinkling effects or back-reaction of $\mathcal{R}$ gradients), and Eq. (17) must currently be read as an empirical fitting function pending derivation.

6. **The Bullet Cluster.** Any resolution field sourced solely by baryonic matter faces the same objection as baryon-sourced modified gravity: in merging clusters the weak-lensing mass peaks separate from the dominant baryonic component (the X-ray gas) and track the galaxies (Clowe et al., 2006). Explaining this requires a dynamical rule for how collisions affect link counting differently for gas and galaxies – no such rule is currently formulated. Until it is, the framework has no account of merging-cluster lensing.

7. Quantum mechanics is not addressed.

8.3 What Would Falsify This Framework

A framework whose central variable parameterizes the whole cosmic history risks becoming unfalsifiable by reinterpretation. To guard against this, we state in advance the observations that would refute the framework *as formulated in this paper*, without appeal to new parameters or post-hoc modifications:

1. **DESI×Planck void stacking at $z \sim 0.3$ yields $\Delta T \approx 2\ \mu$K.** The flagship prediction of Section 7 is $\approx 14\ \mu$K, a factor $\sim$7 above ΛCDM. A stacking measurement consistent with ΛCDM at this redshift refutes the variable-$\mathcal{R}$ account of the ISW anomaly. This test is feasible with existing data.

2. **A confirmed phantom equation of state, $w_0 < -1$.** Postulate 2 requires $\mathcal{R}(t)$ to grow monotonically, hence $\beta_0 > 0$ and $w_0 = -1 + 2\beta_0 \geq -1$ with no adjustable freedom. A robust cosmological convergence on $w_0 < -1$ (or a confirmed phantom crossing) falsifies the dark-energy mechanism outright. This is the cleanest falsifier: it depends on no fitted parameter of the framework.

3. **Direct detection of a dark matter particle** with the relic abundance required by ΛCDM. Postulate 5 attributes the missing-mass phenomenology to causal aliasing rather than particles; a confirmed particle detection removes its motivation and the framework's dark-matter sector with it.

4. **An irreconcilable β_0.** Section 3 reports a factor $\sim$2–3 tension between the BAO reconstruction of β_0 ($\approx$ 0.02–0.07) and the value implied by the DESI parametric fit ($\approx$ 0.14). We commit to the direct $\mathcal{R}(t)$ likelihood analysis as the arbiter: if no single monotonic $\mathcal{R}(t)$ fits the combined BAO+SNe data, the quantitative dark-energy sector fails even though the qualitative quadrant ($w_0 > -1$, $w_a < 0$) matches.

5. **Energy-dependent photon dispersion.** The framework as formulated assumes the underlying sprinkling preserves local Lorentz invariance (Bombelli et al., 1987), so photon propagation speed must be independent of energy and of local $\mathcal{R}$ to current experimental precision. Fermi-LAT gamma-ray-burst bounds already constrain linear Lorentz violation beyond the Planck scale; if any future refinement of the render dynamics predicts energy-dependent arrival times, those bounds apply immediately. Conversely, a confirmed detection of Lorentz-violating dispersion would require a fundamental revision, since the fixed Minkowski canvas (Postulate 1) provides no mechanism for it.

Each rescue of the framework from one of these outcomes would have to generate new testable predictions of its own; rescues that merely reinterpret a failure are not admissible. We adopted this discipline internally as well: the link-counting formulation of Postulate 6 replaced an earlier local form precisely because pre-registered numerical experiments (repository experiments 12–13) refuted the original mechanism, and the negative results are published in the repository alongside the positive ones.

9 Conclusion

We have outlined a single hypothesis – the universe undergoes causal refinement rather than spatial expansion – that gives a common phenomenological language for several independent observational puzzles. The framework is not yet a replacement for ΛCDM; its decisive next step is to derive $\mathcal{R}(t, \mathbf{x})$ from a causal-set action and confront the resulting model with full likelihood analyses. A near-term test at $z \sim 0.3$ with DESI $\times$ Planck is observationally feasible and would distinguish this framework from ΛCDM by a factor of ~ 7 in signal amplitude.

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Data Availability

All data and code are publicly available at `https://github.com/Editorenbici/rendering-universe`.

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